Constrained LinUCB for Crop Recommendation under Agronomic Feasibility Rules

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Abstract

We frame crop recommendation as a contextual bandit over soil and climate readings, with the recommendable crop as the arm and agreement with an agronomist's logged choice as the simulated reward. A disjoint linear LinUCB model (Li et al., 2010) is restricted at every round to the subset of crops whose agronomic feasible range, estimated from training data, contains the current context – turning plain LinUCB into *constrained* LinUCB. We measure both generalisation (held-out match rate and regret) and online learning speed, and find that the agronomic constraint sharply reduces the exploration cost of learning, at a small, measurable risk of excluding the true crop when a test context is an outlier relative to its own training distribution.

1 Introduction

Public crop-recommendation notebooks almost universally treat the problem as multi-class classification (random forests, gradient boosting). This project instead frames it as a *contextual bandit*: a decision made under a single observed context with partial feedback, which is the realistic online setting for a system deployed to make one recommendation at a time and learn from whether it was accepted. LinUCB [1] is a classical, closed-form linear bandit with no gradient descent and no neural network; we add agronomic domain knowledge as a hard constraint on the arm set offered at each round, which is both a safety mechanism and, as Section 6 shows, a substantial accelerant of learning.

2 Dataset

The Crop Recommendation dataset (Kaggle ref `atharvaingle/crop-recommendation-dataset`) contains soil nutrient levels (N, P, K), temperature, humidity, pH, and rainfall, each row labelled with the crop an agronomist recommends for those conditions, across 22 crop types. Following the standard technique for turning a labelled classification dataset into an offline contextual-bandit replay log [2], each row becomes one round: the context is the soil/climate vector, and the simulated reward for choosing arm a is 1 iff a equals the row's logged crop.

3 Feature engineering

Features are declared in `conf/features.yaml`. `context_zscore` standardises the seven raw soil/climate features, fitted strictly on the training replay stream, since LinUCB's ridge prior assumes a roughly isotropic scale. `arm_feasible_range` computes, per crop, the training-time observed (min, max) of every raw feature expanded by an 8% tolerance margin – this is the agronomic constraint table. `arm_popularity` and `arm_context_centroid` are training-only descriptive statistics used by the popularity baseline and reported for interpretability. A registry check asserts the YAML specification and the Python implementation agree in both directions.

4 Methodology

4.1 LinUCB

For each arm a, LinUCB maintains a ridge-regression posterior

$$A_a = I_d + \sum_{t:\, a_t = a} x_t x_t^\top, \qquad b_a = \sum_{t:\, a_t = a} r_t x_t, \tag{1}$$

with point estimate $\theta_a = A_a^{-1} b_a$. The upper-confidence-bound score for context x under arm a is

$$p_a(x) = x^\top \theta_a + \alpha \sqrt{x^\top A_a^{-1} x}, \tag{2}$$

where the second term is an exploration bonus proportional to the model's remaining uncertainty about x under arm a's posterior, and α controls the explore/exploit trade-off. Both A_a and b_a are updated in closed form after every round; no gradient descent is involved.

4.2 Agronomic constraint

Let $R_a = \{(\ell_{a,j}, u_{a,j})\}_{j=1}^{7}$ be crop a's training-time feasible range (Section 3) for each raw feature j. At round t with raw context $\tilde{x}_t$, the feasible arm set is

$$\mathcal{F}_t = \{a : \ell_{a,j} \le \tilde{x}_{t,j} \le u_{a,j} \ \forall j\}. \quad (3)$$

Constrained LinUCB selects $\arg\max_{a \in \mathcal{F}_t} p_a(x_t)$; plain LinUCB selects over every arm. If $\mathcal{F}_t = \emptyset$, every arm is treated as feasible rather than leaving the policy with no action.

5 Experimental protocol

Rounds are shuffled once and split into training (70%), validation (15%) and test (15%). The training partition is replayed in order as an online learning stream: the policy selects an arm, observes the simulated reward, and updates, round by round. Validation/test partitions are then replayed with the policy frozen (no further updates), reporting match rate (the simulated reward rate) and regret (1 − match rate) against an oracle that always matches. We compare constrained LinUCB, plain LinUCB, a context-blind random policy, and an epsilon-greedy popularity baseline, and separately report the online learning curve (mean reward over the first 25, 100, and all training rounds) for constrained versus plain LinUCB, which isolates how much the constraint reduces exploration cost.

6 Results

Table 1: Held-out generalisation on the synthetic CI fixture (test partition).

Model	Match rate	Regret	Mean pool size
ConstrainedLinUCB	0.9556	0.0444	1.73
LinUCB	0.9926	0.0074	8.00
Random	0.8519	0.1481	1.73
EpsilonGreedy	0.9185	0.0815	1.73

Table 2: Online learning curve: mean training reward by round window.

Model	First 25	First 100	All rounds
ConstrainedLinUCB	1.00	0.96	0.978
LinUCB	0.20	0.40	0.870

Table 1 reports held-out generalisation on the synthetic CI fixture. Table 2 reports the online learning curve. Constrained LinUCB reaches a mean training reward of 1.00 within the first 25 rounds, while plain LinUCB – forced to explore all eight arms before its per-arm posteriors converge – needs the full training stream to approach the same level. This is the central result: restricting the arm set with agronomic domain knowledge removes nearly all of the exploration cost. The trade-off is visible in the held-out match rate, where constrained LinUCB is marginally *below* plain LinUCB (0.956 vs 0.993): occasionally a test round's context falls just outside its own crop's training-time feasible range, and the hard constraint excludes the correct answer outright. This is the known risk of hard constraints in constrained bandits, and is reported here rather than hidden.

7 Limitations and conclusion

The simulated-reward protocol assumes every training row's logged crop is the single correct answer, which discards any ambiguity in genuinely multi-suitable soil/climate conditions. The feasible-range tolerance (8%) is a fixed hyperparameter, not learned or validated against an agronomic source; a production system would consult domain experts or a larger reference table instead. The drug-recommendation project elsewhere in this portfolio similarly layers a hard safety constraint over a statistical model and warns against over-reliance on it; the same caution applies here; crop recommendations should support an agronomist's judgement, not replace it. Overall, constrained LinUCB demonstrates a clear, measurable benefit (learning speed) and a clear, measurable cost (occasional exclusion of the correct arm) from the same mechanism, which is a more honest result than either in isolation.

References

[1] Lihong Li, Wei Chu, John Langford, and Robert E. Schapire. A contextual-bandit approach to personalized news article recommendation. In *Proceedings of the 19th International Conference on World Wide Web*, pages 661–670, 2010.

[2] Lihong Li, Wei Chu, John Langford, and Xuanhui Wang. Unbiased offline evaluation of contextual-bandit-based news article recommendation algorithms. In *Proceedings of the Fourth ACM International Conference on Web Search and Data Mining*, pages 297–306, 2011.